Adaptive ISPG in LS-Dyna

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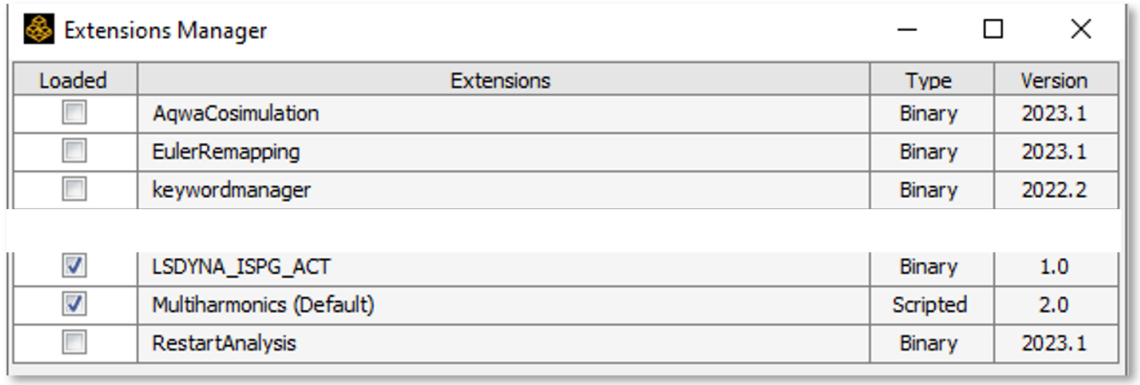
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Adaptive ISPG ACT App in WB LS-Dyna

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Installing the extension

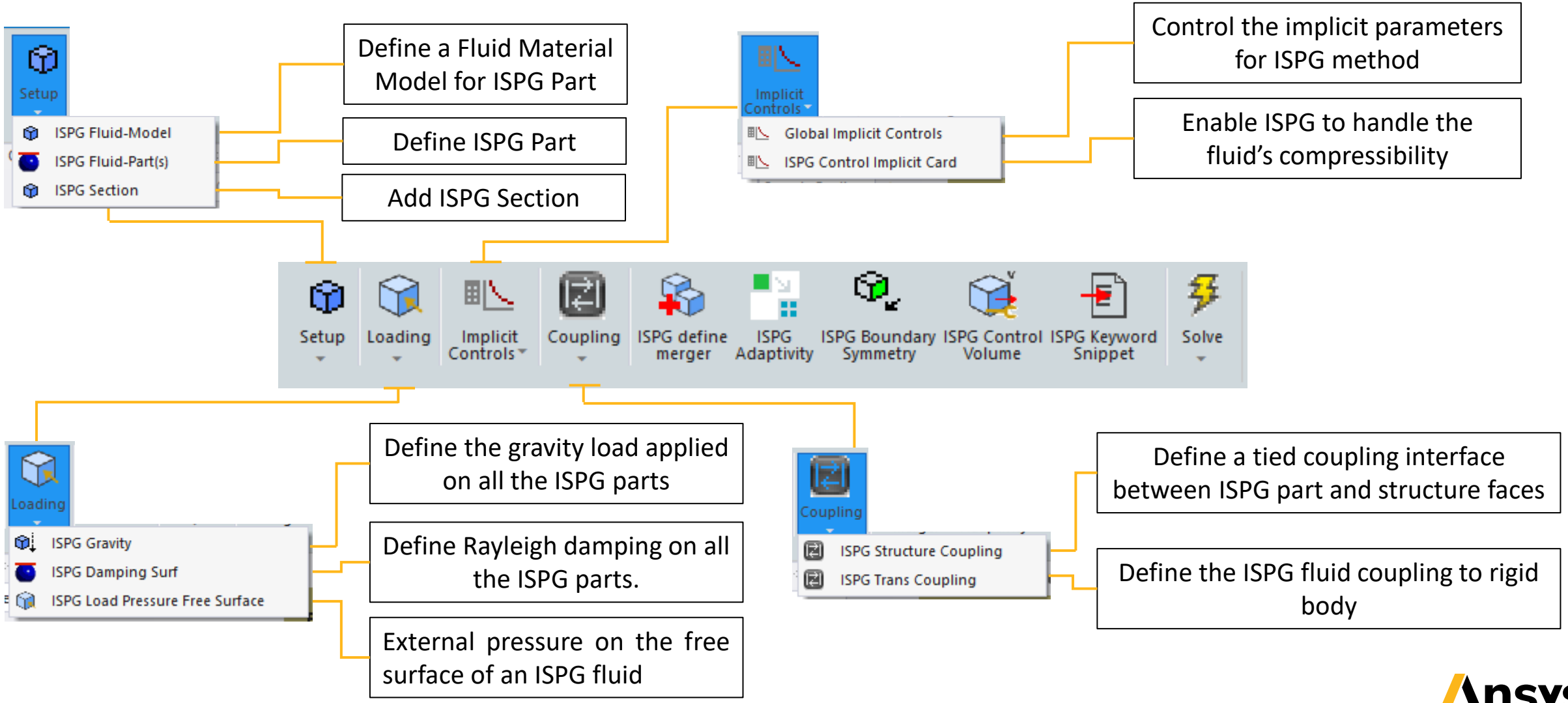
- LSDYNA_ISPG_ACT extension requires at least version 2024R1.
- Install the binary from WB Project page – Extensions > Install Extension > LSDYNA_ISPG_ACT.wbex



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Adaptive ISPG Extension in Mechanical

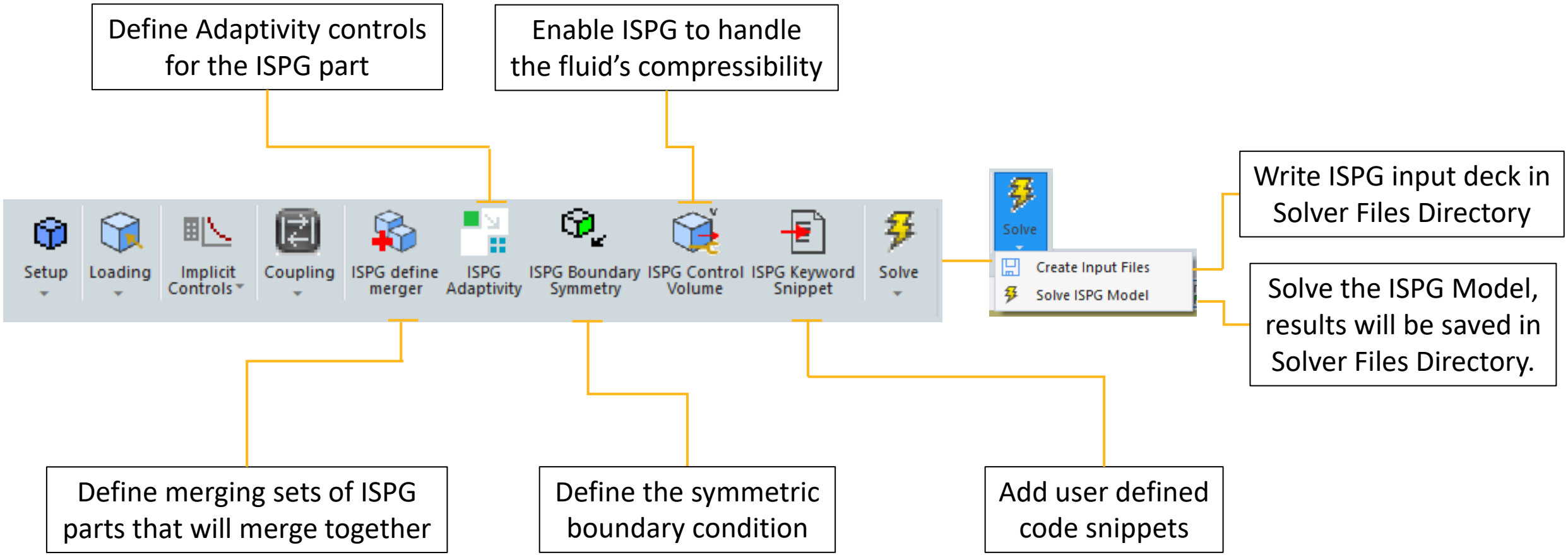
- Adaptive ISPG Ribbon in Mechanical:



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Adaptive ISPG Extension in Mechanical

- Adaptive ISPG Ribbon in Mechanical:



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ACT objects to LS-Dyna Keywords Map

Gravity load on parts (ACT)



Details of "ISPG Gravity"

Definition	
Magnitude	
Global Direction	Global Y

Change the defaults if needed

NOTE : The decimal values will be rounded off to 5 digits after the decimal point **for all cards**.
e.g., 1.123456789 will be rounded off to 1.12346

LS-Dyna Keywords

***ISPG_LOAD_GRAVITY**

Purpose: define the gravity load applied on all the ISPG parts.

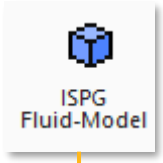
Card 1	1	2	3	4	5	6	7	8
Variable	DOF	GRA						
Type	I	F						
Default	None	none						

DOF	Global axis direction for the symmetric boundary condition DOF=1, global x axis DOF=2, global y axis DOF=3, global z axis
GRA	Value of the gravity load

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ACT objects to LS-Dyna Keywords Map

Fluid Material Model (ACT) – Newtonian



Details of "ISPG Fluid-Material Model"

Definition	
Material ID	8
Fluid Type	Newtonian Fluid Model
Temperature dependency	No
Density	
Surface Tension	
Viscosity	

Details of "ISPG Fluid-Material Model"

Definition	
Material ID	8
Fluid Type	Newtonian Fluid Model
Temperature dependency	Yes
Alpha	
Tref	
Density	
Surface Tension	
Viscosity	

LS-Dyna Keywords

*MAT_ISPG_ISO_NEWTONIAN

***MAT_ISPG_ISO_NEWTONIAN**

This material type is used specifically for modeling the Newtonian flow behavior of incompressible free surface flow with incompressible SPG. It was developed to predict the shape evolution of solder joints during the electronic reflow process. See Pan et al 2020 for details.

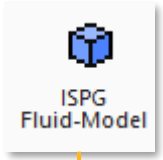
Card 1	1	2	3	4	5	6	7	8
Variable	MID	RO	DYNVIS	SFTEN				
Type	A	F	F	F				

Card 2	1	2	3	4	5	6	7	8
Variable	ALPHA	TREF						
Type	F	F						

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ACT objects to LS-Dyna Keywords Map

Fluid Material Model (ACT) – Carreau



Definition	
Material ID	8
Fluid Type	Non-Newtonian (Carreau) Fluid Model
Temperature dependency	No
Density	
Surface Tension	
Viscosity @ Zero Shear Rate	
Viscosity @ Infinity Shear Rate	
Time Constant	0.3
Power Law Index	0.5

Details of "ISPG Fluid-Material Model"

Definition	
Material ID	8
Fluid Type	Non-Newtonian (Carreau) Fluid Model
Temperature dependency	Yes
Alpha	
Tref	
Density	
Surface Tension	
Viscosity @ Zero Shear Rate	
Viscosity @ Infinity Shear Rate	
Time Constant	0.3
Power Law Index	0.5

LS-Dyna Keywords

*MAT_ISPG_CARREAU

***MAT_ISPG_CARREAU**

The Carreau model attempts to describe a wide range of fluids by the establishment of a curve-fit to piece together functions for both Newtonian and shear-thinning(n<1) non-Newtonian law.

Card 1	1	2	3	4	5	6	7	8
Variable	MID	RO	VISC0	SFTEN	VISC_LIM	LAMBDA	n	
Type	A	F	F	F				

Card 2	1	2	3	4	5	6	7	8
Variable	ALPHA	TREF						
Type	F	F						

In the Carreau model, the viscosity is described as:

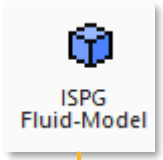
$$\eta = \eta_\infty + (\eta_0 - \eta_\infty)(1 + \gamma^2\lambda^2)^{(n-1)/2}$$

Where λ is time constant, n is the power-law index, η_0 and η_∞ are the zero- and infinite-shear viscosities respectively.

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ACT objects to LS-Dyna Keywords Map

Fluid Material Model (ACT) – Cross Model



Details of "ISPG Fluid-Material Model"

Definition	
Material ID	8
Fluid Type	Cross Model
Temperature dependency	No
Density	
Surface Tension	
Viscosity @ Zero Shear Rate	
Time Constant	0.3
Power Law Index	0.5

Details of "ISPG Fluid-Material Model"

Definition	
Material ID	8
Fluid Type	Cross Model
Temperature dependency	Yes
Alpha	
Tref	
Density	
Surface Tension	
Viscosity @ Zero Shear Rate	
Time Constant	0.3
Power Law Index	0.5

LS-Dyna Keywords

***MAT_ISPG_CROSSMODEL**

The Cross model attempts to describe the shear rate dependence across the Newtonian region and the shear-thinning region.

Card 1	1	2	3	4	5	6	7	8
Variable	MID	RO	VISC0	SFTEN		LAMBDA	n	
Type	A	F	F	F		F	F	

Card 2	1	2	3	4	5	6	7	8
Variable	ALPHA	TREF						
Type	F	F						

In the Cross model, the viscosity is described as:

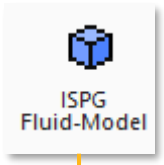
$$\eta = \frac{\eta_0}{1 + (\lambda\dot{\gamma})^{1-n}}$$

Where $\dot{\gamma} = \sqrt{\frac{1}{2}\overline{\overline{D}}:\overline{\overline{D}}}$ is the second invariant of the rate-of-deformation tensor $\overline{\overline{D}} = \left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial j}\right)$, λ is time constant, n is the power-law index, η_0 is the zero-shear-rate viscosities.

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ACT objects to LS-Dyna Keywords Map

Fluid Material Model (ACT) – Iso Non-Newtonian



Details of "ISPG Fluid-Material Model"

Definition	
Material ID	8
Fluid Type	Iso Non-newtonian Model
Temperature dependency	No
Density	
Surface Tension	
Load Curve ID	

Details of "ISPG Fluid-Material Model"

Definition	
Material ID	8
Fluid Type	Iso Non-newtonian Model
Temperature dependency	Yes
Alpha	
Tref	
Density	
Surface Tension	
Load Curve ID	

LS-Dyna Keywords

***MAT_ISPG_ISO_NONNEWTONIAN**

This material type is specifically used for modeling the non-Newtonian flow behavior of incompressible free surface flows with incompressible SPG. The Load Curve ID (LCID) is necessary to define the relationship between viscosity and shear rate.

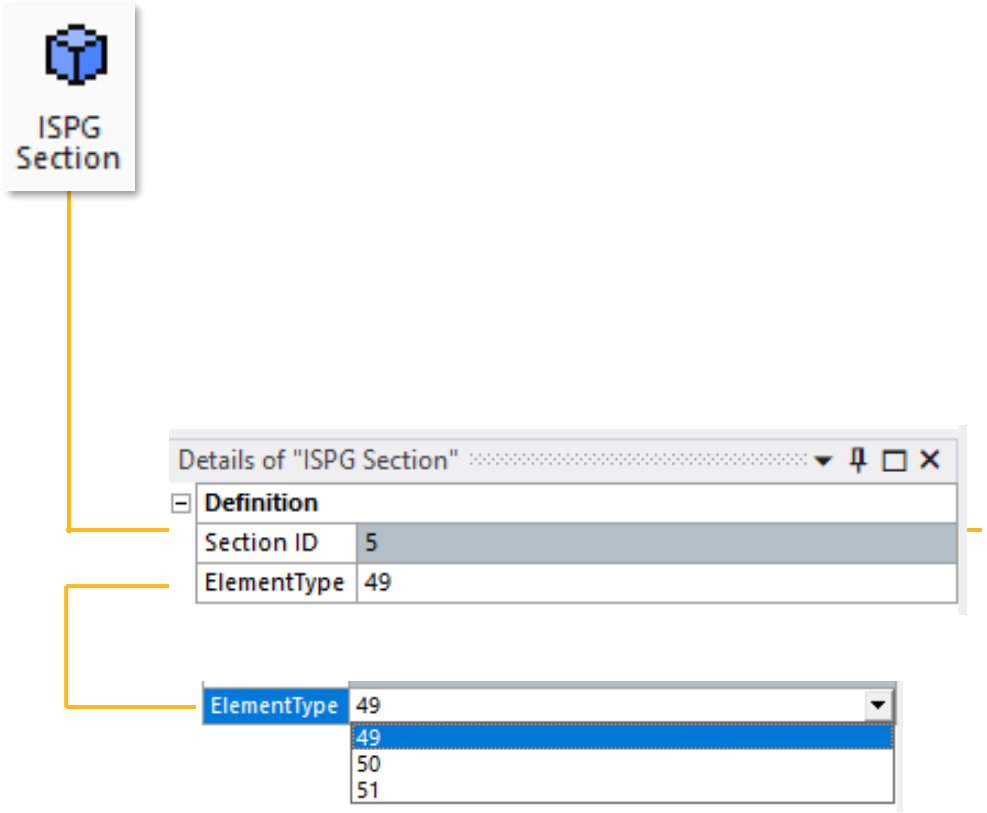
Card 1	1	2	3	4	5	6	7	8
Variable	MID	RO	LCID	SFTEN				
Type	A	F	I	F				

Card 2	1	2	3	4	5	6	7	8
Variable	ALPHA	TREF						
Type	F	F						

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ACT objects to LS-Dyna Keywords Map

Fluid Material Model (ACT) - CARD 2



LS-Dyna Keywords

***SECTION_ISPG**

Purpose: Define section properties for incompressible smoothed particle Galerkin (ISPG) element.

Card 1	1	2	3	4	5	6	7	8
Variable	SECID	ELFORM						
Type	I/A	I						

ELFORM	Element formulation options. EQ.49: 3D and pseudo 2D ISPG formulation. EQ.50: Planar 2D ISPG formulation. EQ.51: Axisymmetric 2D ISPG formulation.

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ACT objects to LS-Dyna Keywords Map

Fluid Material Model (ACT)



Select / Scope Part(s) from GUI

Geometry	
Scoping Method	Geometry Selection
Geometry	
Definition	
Material ID	
Section ID	

Add Material and Section ID defined in CARD 1 and CARD 2

LS-Dyna Keywords

***PART**

Card 1	1	2	3
Variable	PID	SECID	MID
Type	I	I	I

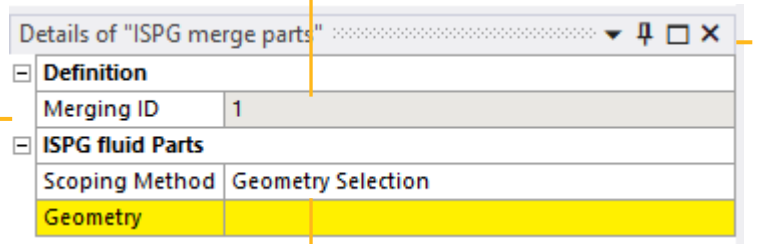
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ACT objects to LS-Dyna Keywords Map

Define Merger MPP(ACT)



Merging ID
(Defined by default)



Select the ISPG fluid Parts that might merge if close enough

LS-Dyna Keywords

Card 1	1	2	3	4	5	6	7	8
Variable	MID	PID	PID	PID	PID	PID	PID	PID
Type	I	I	I	I	I	I	I	I
Default	none	none	none	none	none	none	none	none

MID	Merging ID (all the ISPG parts in this ID will be distributed into the same processor during the MPP run).
PID	ISPG fluid Part ID.

NOTE: Maximum 7 bodies can be selected. If more bodies need to be selected, add another 'ISPG merge parts' object or use 'ISPG Keyword Snippet'.

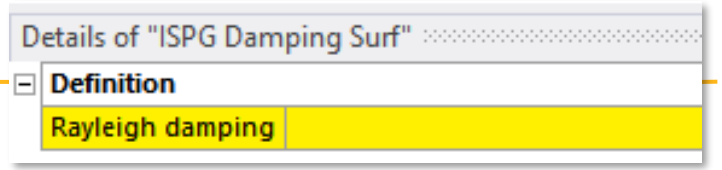
(Supported only for 24R2 and LS-DYNA dev version)

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ACT objects to LS-Dyna Keywords Map

Rayleigh Damping Value(ACT)





LS-Dyna Keywords

***ISPG_ DAMPING_SURF**

Purpose: define Rayleigh damping related to fluid surface tension on all the ISPG parts. Currently, this feature supports only 2D ISPG formulation.

Card 1	1	2	3	4	5	6	7	8
Variable	DAMP							
Type	F							
Default	None							

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ACT objects to LS-Dyna Keywords Map

ISPG Load Pressure Free Surface(ACT)



Details of "ISPG Load Pressure Free Surface"

Definition	
Applied pressure	
Time to activate applied pressure	
Time to deactivate applied pressure	
Minimum x-coordinate	
Maximum x-coordinate	
Minimum y-coordinate	
Maximum y-coordinate	
Minimum z-coordinate	
Maximum z-coordinate	

LS-Dyna Keywords

*ISPG_LOAD_PRESSURE_FREE_SURFACE

*ISPG_LOAD_PRESSURE_FREE_SURFACE

Purpose: To define the application of external pressure on the free surface of an ISPG fluid within a specified box-shaped volume.

Card 1 defines the external pressure applied to the free surface of the ISPG fluid.

Card 2 specifies a box-shaped volume by providing the global coordinates of two diagonally opposite corner points. This box volume determines the free surface area where the external pressure is applied.

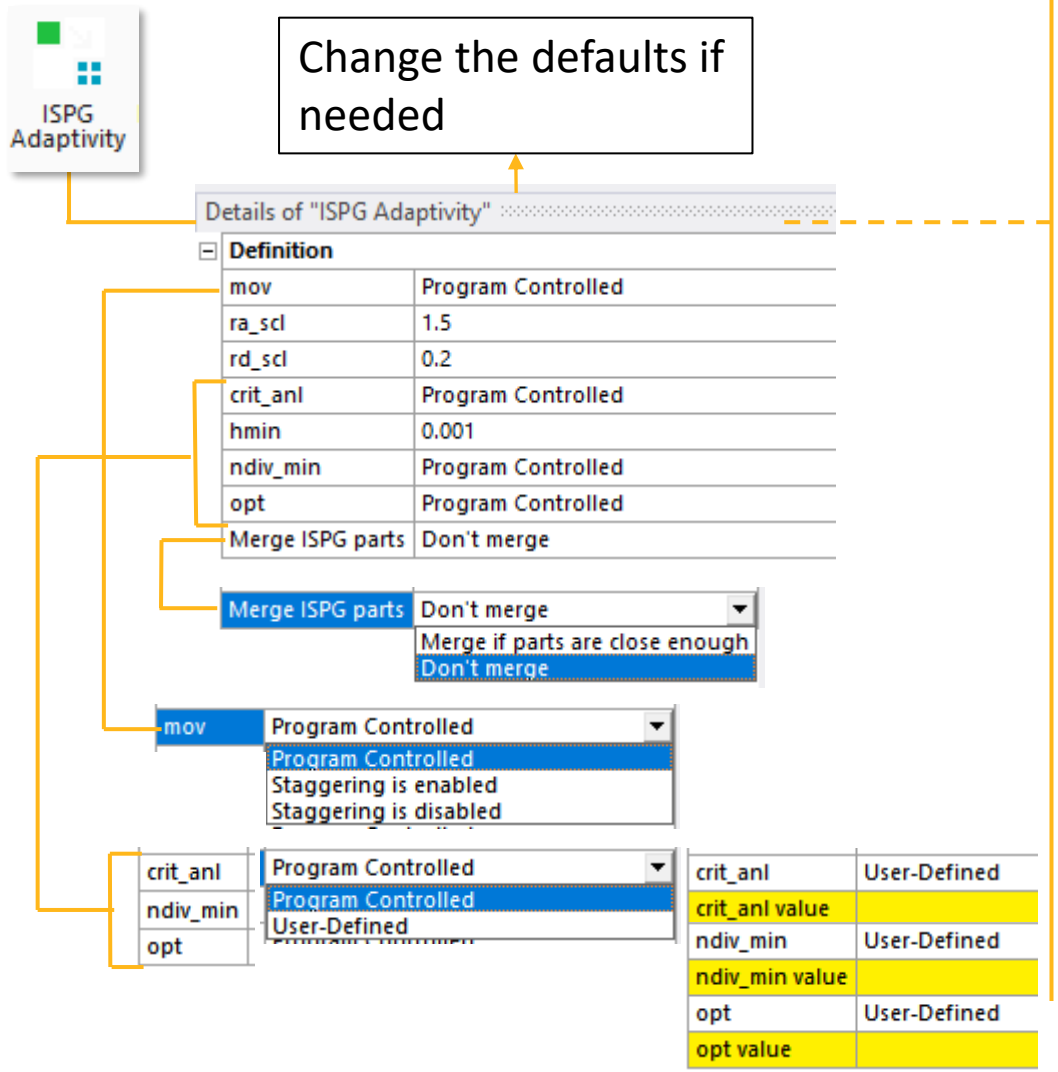
Card 1	1	2	3	4	5	6	7	8
Variable	PE	BIRTH	DEATH					
Type	F	F	F					

Card 2	1	2	3	4	5	6	7	8
Variable	XMN	XMX	YMN	YMX	ZMN	ZMX		
Type	F	F	F	F	F	F		

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ACT objects to LS-Dyna Keywords Map

ISPG Adaptivity (ACT)



LS-Dyna Keywords

*ISPG_ CONTROL_ADAPTIVITY

2	3	4	5	7	8
MOV	RA_SCL	RD_SCL	CRIT_ANL	NDIV_MIN	OPT
I	F	F	F	I	I
0	1.5	0.2	0.524	0	0

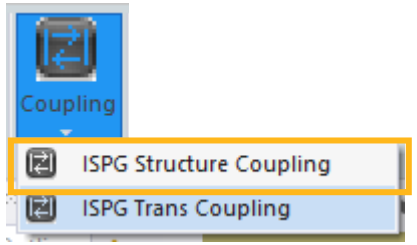
Card 2	1	3
Variable	HMIN	MERG
Type	F	I
Default	0.001	0

MOV	EQ. 0 or 1: staggering is enabled. Otherwise: staggering is not disabled.
RA_SCL	Criteria for adding a new ISPG node if the distance of two adjacent ISPG nodes is larger than RA_SCL × D_{avg}, here D_{avg} is the original averaged nodal distance of an ISPG part.
RD_SCL	Criteria for deleting an ISPG node if the distance of two adjacent ISPG nodes is smaller than RD_SCL × D_{avg}, here D_{avg} is the original averaged nodal distance of an ISPG part.
CRIT_ANL	Critical angle (unit rad) of adjacent structural segment for ISPG edges staggering (see MOV defined above).
NDIV_MIN	The minimum number of divisions of the structural segment for ISPG nodes alignment (see ALIGN defined in above).
OPT	OPT is used to control the number of the quadrature points for the detection of the ISPG element into the solid ones.
HMIN	Minimum relative element height compared to the average nodal distance D_{avg}.
MERG	Flag to automatically merge ISPG parts into one part if those parts are close enough.

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ACT objects to LS-Dyna Keywords Map

ISPG Structure Coupling (ACT)





Details of "ISPG Structure Coupling"

Definition	
Coupling ID	2
Static (equilibrium) contact angle	
Advancing contact angle	Program Controlled
Receding contact angle	Program Controlled
Coupling output time-step	
Type of boundary condition for contact edges (sbc)	Contact edges can move on structure surface
Scaling factor	0.5
Slip length for Navier slip model	0 m
Target faces scoping	
Scoping Method	Geometry Selection
Geometry	
ISPG Part	
Scoping Method	Geometry Selection
Geometry	

ISPG Part: Select / Scope Part(s) from GUI
Target Faces scoping: Select / Scope faces from GUI
More information on next slide.

EQ.0: Contact edges can move on structure surface.
EQ.1: Contact edges tie with structure surface.
EQ.2: Free-slip boundary condition.

LS-Dyna Keywords

***DEFINE_ISPG_TO_SURFACE_COUPLING**

Purpose: Define a coupling interface between fluid particles modeled with implicit smoothed particle Galerkin (ISPG) and a surface.

Card 1	1	2	3	4	5	6	7	8
Variable	FP	SURF	FPTYPE	SURFTYPE	CPL_ID			
Type	I	I	I	I	I			
Default	none	none	none	none	none			

Card 2	1	2	3	4	5	6	7	8
Variable	SBC	SCA	ACA	RCA	VCA	SFPN	THK	LSLP
Type	I	F	F	F	F	F	F	F
Default	0	none	none	non	1.0	1.0	0.5	0.0

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ACT objects to LS-Dyna Keywords Map

ISPG Structure Coupling (ACT)

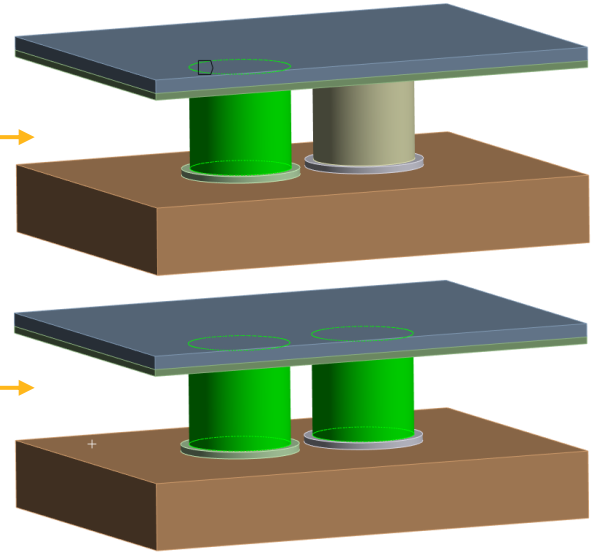
ISPG Part scoping:

- You can select one or more geometries (Body) simultaneously as single ISPG Part scoping.
- Alternatively, you can create Named Selection with one or more bodies and scope it.

ISPG Part	
Scoping Method	Geometry Selection
Geometry	1 Body

ISPG Part	
Scoping Method	Geometry Selection
Geometry	2 Bodies

ISPG Part	
Scoping Method	Named Selection
Named Selection	Selection_Parts



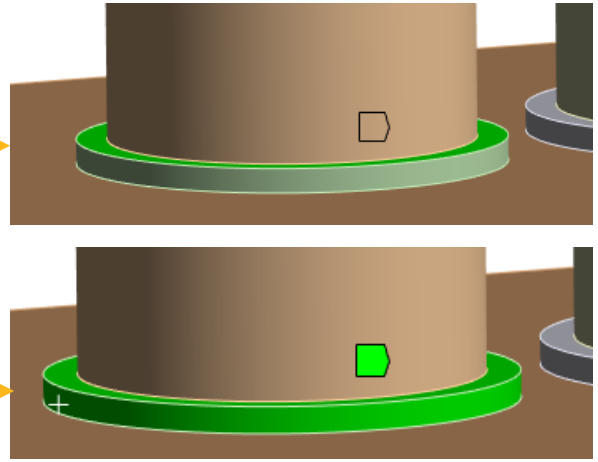
Target Faces scoping:

- You can select one or more geometries (Faces) simultaneously as single Target Faces scoping.
- Alternatively, you can create Named Selection with one or more faces and scope it.

Target faces scoping	
Scoping Method	Geometry Selection
Geometry	1 Face

Target faces scoping	
Scoping Method	Geometry Selection
Geometry	2 Faces

Target faces scoping	
Scoping Method	Named Selection
Named Selection	Selection_Faces

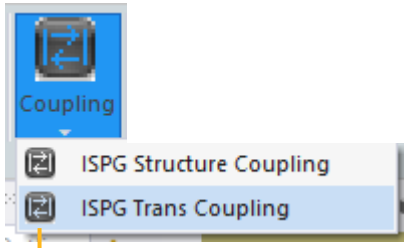


Note: After creating a Named Selection for ISPG Structure Coupling, if you don't set the 'LS-DYNA Named Selection User Id', the extension will set this value.

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ACT objects to LS-Dyna Keywords Map

ISPG_TRANS_CPL



Details of "ISPG Trans Coupling 2"

Trans cpl Parts	
Scoping Method	Geometry Selection
Geometry	
Definition	
Translational X Degree of Freedom	Not fixed
Translational Y Degree of Freedom	Not fixed
Translational Z Degree of Freedom	Not fixed
Rotational X Degree of Freedom	Not fixed
Rotational Y Degree of Freedom	Not fixed
Rotational Z Degree of Freedom	Not fixed
Time to activate the strong coupling algorithm	0
Minimum thickness of the fluid under the compression of rigid bodies	0

LS-Dyna Keywords

Card 1	1	2	3	4	5	6	7	8
Variable	PID1	PID2	PID3	PID4	PID5	PID6	PID7	PID8
Type	I	I	I	I	I	I	I	I
Default	none	none	none	none	none	none	none	non

Card 2	1	2	3	4	5	6	7	8
Variable	DOFX	DOFY	DOFZ	DOFRX	DOFRY	DOFRZ	BIRTH	THK_MIN
Type	I	I	I	I	I	I	F	F
Default	0	0	0	0	0	0	0.0	0.0

VARIABLE	DESCRIPTION
PID1-8	The parts used for the strong coupling.
DOFX, DOFY, DOFZ	If translational degree of freedom is fixed for the rigid body motion: EQ 0: not fixed, EQ 1: fixed
DOFRX, DOFRY, DOFRZ	If rotational degree of freedom is fixed for the rigid body motion: EQ 0: not fixed, EQ 1: fixed
BIRTH	The birth time is used to activate the strong coupling algorithm.
THICK_MIN	Variable used to define the minimum thickness of the fluid under the compression of rigid bodies.

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ACT objects to LS-Dyna Keywords Map

ISPG Implicit Controls (ACT)



Definition	
Initial Time Step	
Minimum Time Step	
Maximum Time Step	

Currently, only three parameters can be adjusted, in a long list of default implicit controls.

LS-Dyna Keywords

```
*CONTROL_IMPLICIT_AUTO
$#    iauto     iteopt     itewin      dtmin      dtmax      dtexp     kfail     kcycle
         1         11          5      0.002        0.2        0.0         0         0
*CONTROL_IMPLICIT_DYNAMICS
$#    imass      gamma       beta     tdybir     tdydth     tdybur     irate      alpha
         1        0.6       0.38        0.01.00000E281.00000E28         0        0.0
*CONTROL_IMPLICIT_GENERAL
$#  imflag        dt0     imform       nsbs        igs      cnstn       form     zero_v
         1       0.02          2          1          2          0          0          0
*CONTROL_IMPLICIT_SOLUTION
$#  nsolvr     ilimit     maxref      dctol      ectol      rctol      lstol     abstol
        12         11         15      0.001       0.011.00000E10        0.91.0000E-10
$#   dnorm     diverg      istif    nlprint    nlnorm   d3itctl     cpchk
         2          1          1          2          0          0         0
$#  arcctl     arcdir     arclen     arcmth     arcdmp     arcpsi     arcalf     arctim
         0          0        0.0          1          2          0         0          0
$#   lsmtd      lsdir       irad       srad       awgt       sred
         4          2        0.0        0.0        0.0        0.0
*CONTROL_IMPLICIT_SOLVER
$#  lsolvr     lprint      negev      order       drcm     drcprm    autospc    autotol
        22          0          2          0          4        0.0          1        0.0
$#  lcpack     mtxdmp     iparm1     rparm1     rparm2
         2          0       5001.00000E-9      0.001
$#  emxdmp     rdcmem
         0       0.85
*CONTROL_SOLUTION
$      SOLN        NLQ      ISNAN      LCING
$#     soln        nlq      isnan      lcint      lcacc      ncdcf      nocop
         0          0          0        100          0          1          0
*CONTROL_TERMINATION
$    endtim     endcyc      dtmin     endeng     endmas                    unused
$#   endtim     endcyc      dtmin     endeng     endmas      nosol
     200.0   10000000       0.02        0.01.000000E8          0
$*CONTROL_DYNAMIC_RELAXATION
$    nrcyck      drtol     drfctr     drterm     tssfdr     irelal     edttl     idrflg
$         0          0          0      0.001          0          0         0          5
```


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ACT objects to LS-Dyna Keywords Map

ISPG Control Volume (ACT)



Details of "ISPG Control Volume"

Definition	
Time of starting to model the fluid's compressibility	
Inverse of the fluid's bulk modulus	

LS-Dyna Keywords

***ISPG_ CONTROL_VOLUME**

Purpose: enable ISPG to handle the fluid's compressibility for non-free surface flow problems, e.g., the compression molding as all the space is filled in by the fluids. For non-free surface flow problems, the incompressible algorithm may not work, and the compressibility of fluid must be considered to avoid numerical instability.

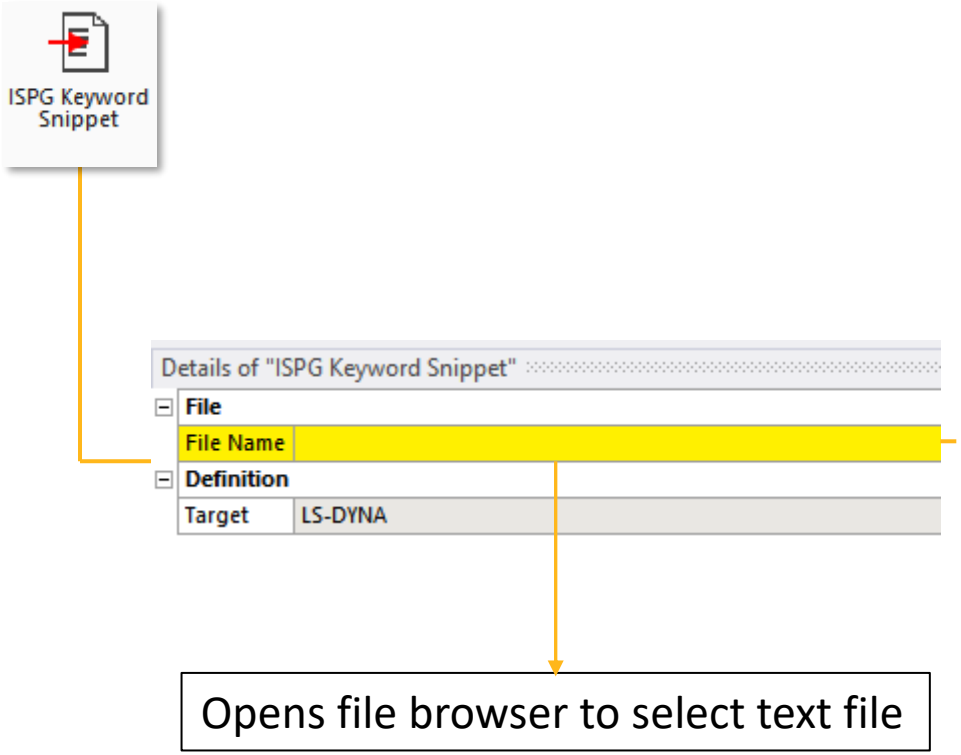
Card 1	1	2	3	4	5	6	7	8
Variable	T	C						
Type	F	F						
Default	None	0.0						

T	Time of starting to model the fluid's compressibility.
C	Inverse of the fluid's bulk modulus

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ACT objects to LS-Dyna Keywords Map

ISPG Keyword Snippet (ACT)



- This button gives users an option to add their own ISPG cards.
- User can select the text file, and that file will be read by the ACT and the contents will be added to ‘include_ispg.k’ file.

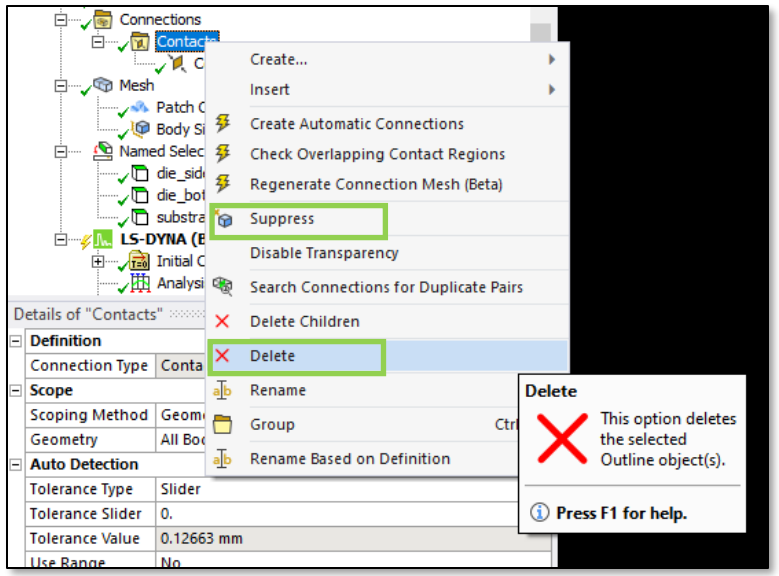
NOTE: This should be used only for ISPG cards. e.g. *ISPG_BOUNDARY_SYMMETRY, *SECTION_ISPG etc.

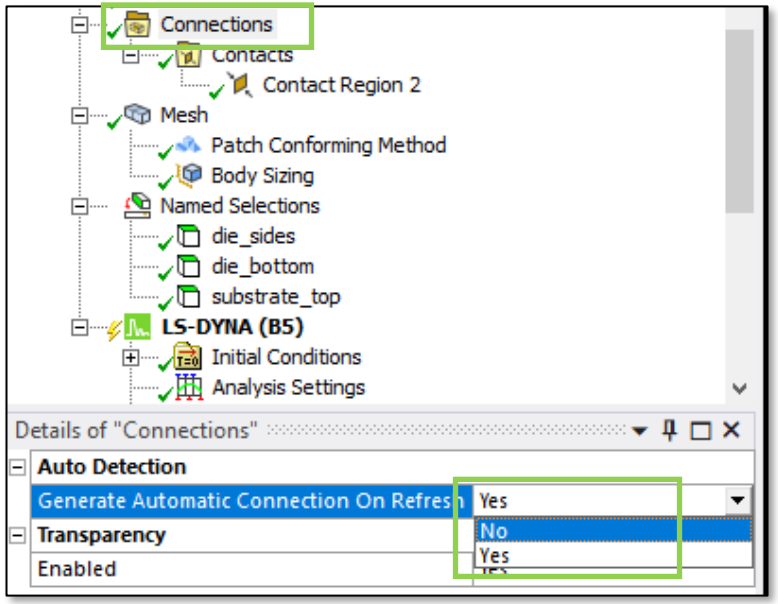
If this is used for other LS-DYNA Cards, LS-Dyna will error terminate. Use Native Keyword objects for that.

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Check Connections

- Make sure you suppress or delete the connections generated automatically by WB LS-Dyna, between the ISPG part and the rest of the model.
- Switch off the automatic contact generation to “No”, to skip auto-generation from next time.

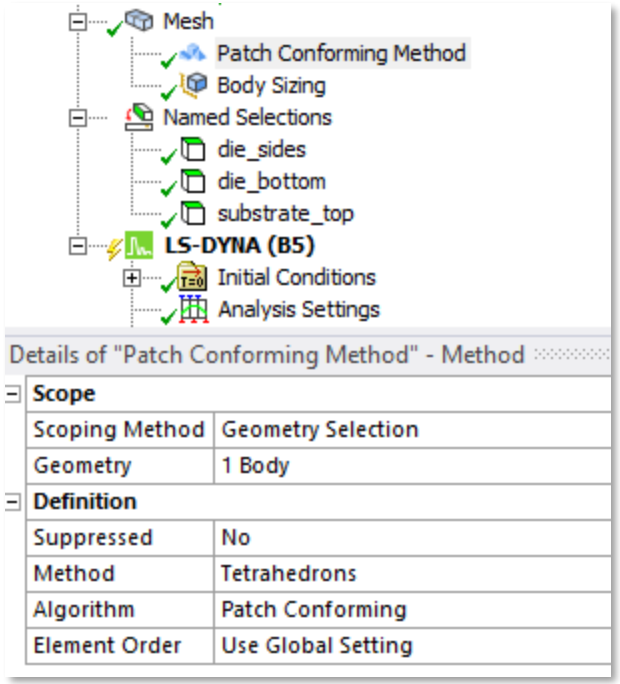




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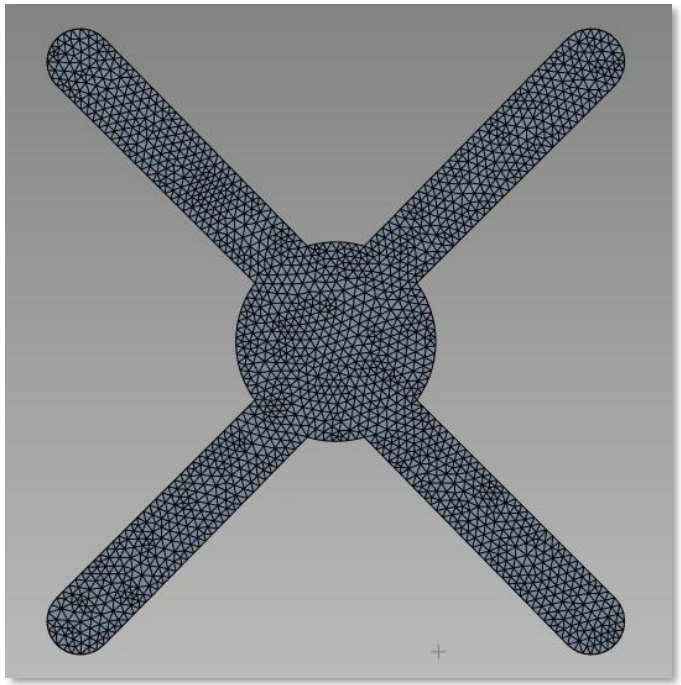
Tetrahedral mesh for ISPG part

- LS-Dyna only accepts tetrahedron elements for the ISPG part.
- Make sure you use the “Patch Conforming Method” to generate a tetrahedral mesh for ISPG part.



Details of "Patch Conforming Method" - Method

Scope	
Scoping Method	Geometry Selection
Geometry	1 Body
Definition	
Suppressed	No
Method	Tetrahedrons
Algorithm	Patch Conforming
Element Order	Use Global Setting



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Writing input deck

- Create Input Files will create an input deck, which you can manually edit (if needed) and run externally, for example with LS-Run. These files are stored in the solver files directory .



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Solving

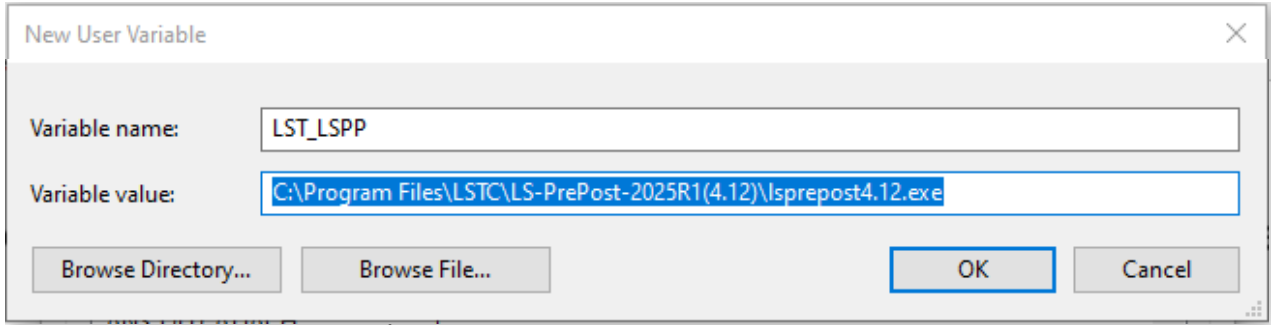
- Solve will start an implicit ISPG run based on the model set-up in WB LS-Dyna, the results will be saved in the solver files directory but will not be loaded back in WB LS-Dyna. User must either manually load in LS-PrePost or use LST_PrePost ACT extension in WB LS-Dyna.
- Solving is only supported for ‘Windows’, and it will not work on ‘Linux’. Linux users can use the App to generate input-deck on Linux and solve outside Mechanical.



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Install LST-PrePost

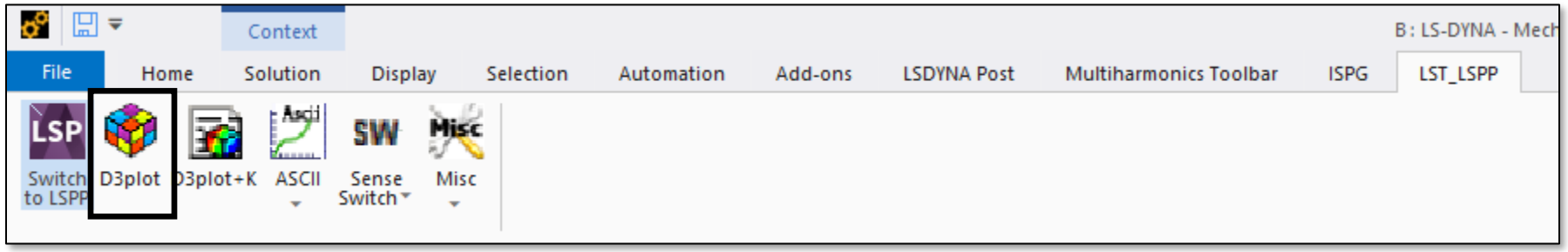
- Install the latest LS-PrePost (> 4.12) – Ansys download page
- Download and install LST_PrePost App from Ansys AppStore
- Set the following Environment Variable '**LST_LSPP**', and value as the installed exe file of LS-PrePost :



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Postprocessing in LS-PrePost

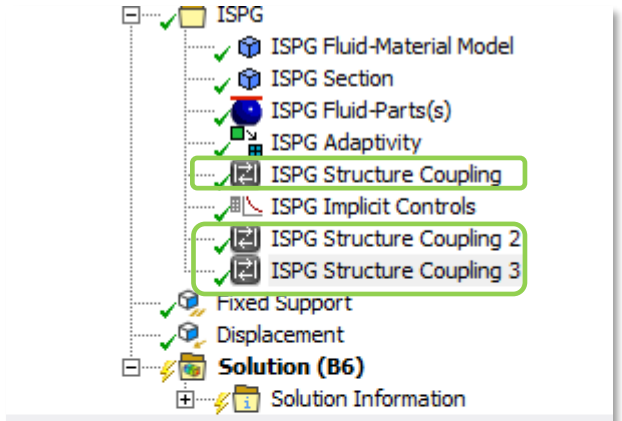
- Once the simulation has reach normal termination, use the LST_LSPP app to open the D3plot in LS-PrePost.
- The app currently does not support D3plot+K functionality.



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ISPG coupling output

- For each defined coupling ID, LS-Dyna will write out an ASCII output file `ispg_cpl`



Details of "ISPG Structure Coupling 3"

Definition	
Coupling ID	4
Static (equilibrium) contact angle	25 °
Coupling output time-step	0.001 s
Target Face(s) LS-DYNA NS ID	103
ISPG Part	
Scoping Method	Geometry Selection
Geometry	1 Body

ispg_cpl

```
$         ISPG to surface coupling time history data
      No. of coupling  =   1

      time =      0.100000E+01
        ID              area                fx                fy                fz
         1      0.000000E+00      0.000000E+00      0.000000E+00      0.000000E+00

      time =      0.509678E+01
        ID              area                fx                fy                fz
         1      0.571889E+01      0.370941E-10      0.543619E-09     -0.915109E-12

      time =      0.153874E+02
        ID              area                fx                fy                fz
         1      0.602697E+01      0.232217E-10     -0.163139E-09     -0.839919E-11

      time =      0.153874E+02
        ID              area                fx                fy                fz
         1      0.602697E+01      0.232217E-10     -0.163139E-09     -0.839919E-11

      time =      0.253874E+02
        ID              area                fx                fy                fz
         1      0.618765E+01     -0.302329E-10      0.228476E-09      0.487148E-11

      time =      0.253874E+02
        ID              area                fx                fy                fz
         1      0.618765E+01     -0.302329E-10      0.228476E-09      0.487148E-11
```

Output:

- *Reaction forces.*
- *Contact Area between the ISPG part and the target face(s).*

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NOTE: Ansys and LS-DYNA Versions

- **We recommend using Ansys 25R2 and LS-DYNA version 16.0.0 for latest capabilities.**
- If the user wants to use old Mechanical versions - use the App only to generate the input deck and then solve with LS-Run (Refer Slide 37 : How to use LS-Run)

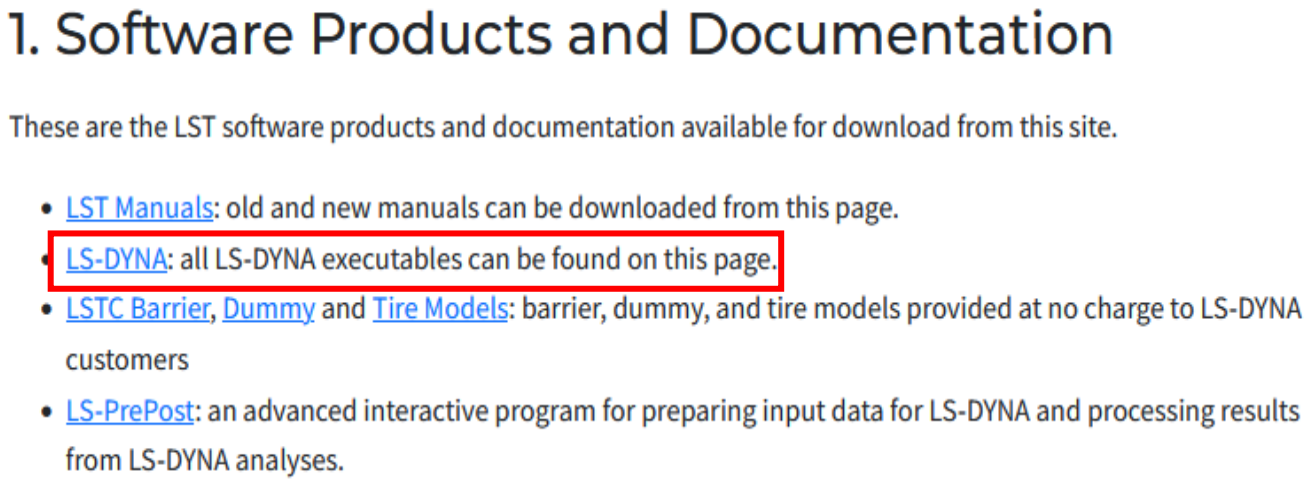
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Download latest LS-DYNA version

- Go to download page of LS-DYNA, or click on the following link:

  https://lsdyna.ansys.com/download-install-overview/

- Click on the link highlighted below to search for latest LS-DYNA executable:



1. Software Products and Documentation

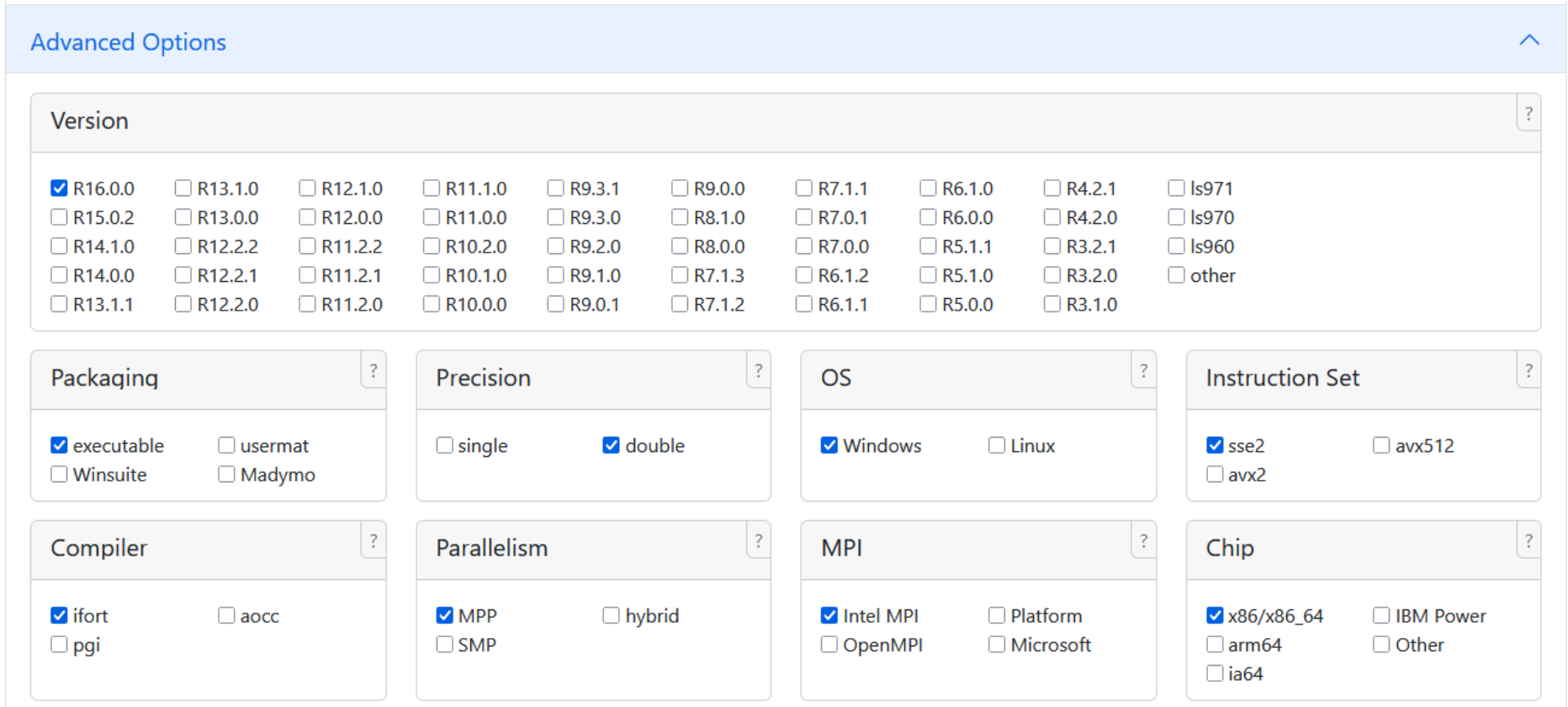
These are the LST software products and documentation available for download from this site.

- LST Manuals: old and new manuals can be downloaded from this page.
- LS-DYNA: all LS-DYNA executables can be found on this page.
- LSTC Barrier, Dummy and Tire Models: barrier, dummy, and tire models provided at no charge to LS-DYNA customers
- LS-PrePost: an advanced interactive program for preparing input data for LS-DYNA and processing results from LS-DYNA analyses.

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Download latest LS-DYNA version

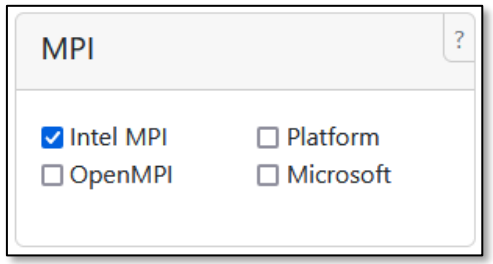
- After landing on the LS-DYNA download page, please select Advanced options and select the version **R16.0.0** and the options as shown below:

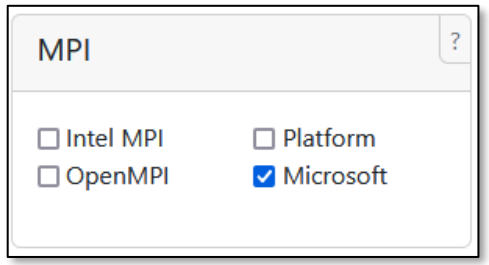


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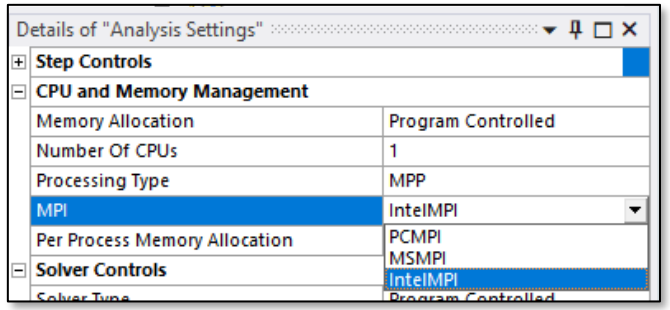
Download latest LS-DYNA version

- The ACT Extension currently supports only MPP double precision LS-Dyna solver.
- User can choose the MPI from IntelMPI or Microsoft MPI. Download the correct LS-DYNA version based on the MPI you are going to use and set in Mechanical.





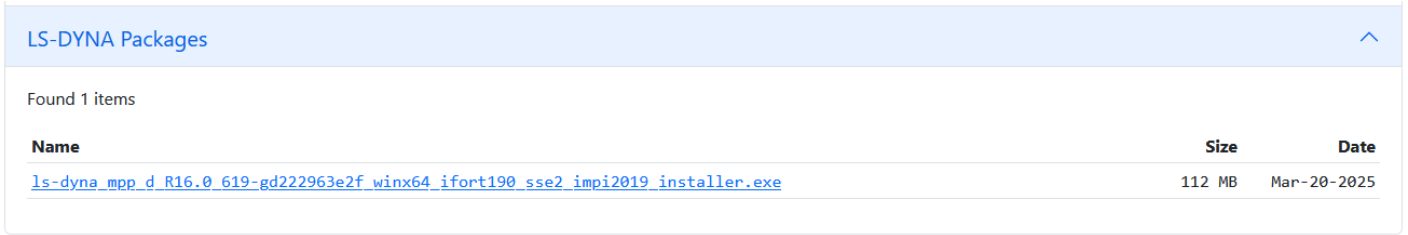
- Please make sure to set the environment values correctly in the next steps.
- Make sure correct settings are applied in the analysis settings in Mechanical while setting up the simulation.



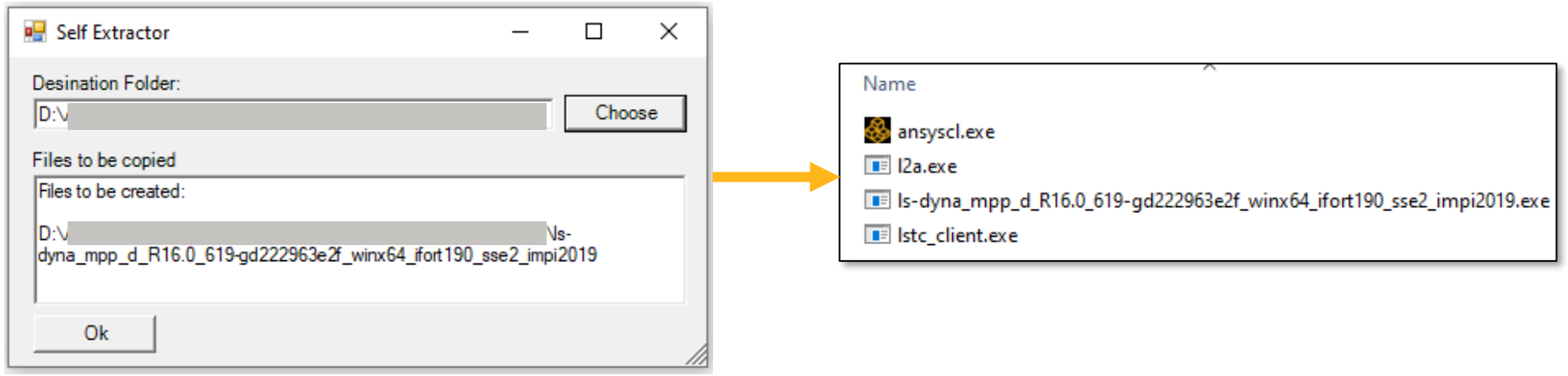
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Download latest LS-DYNA version

- This will give the available LS-DYNA version for download as shown below:



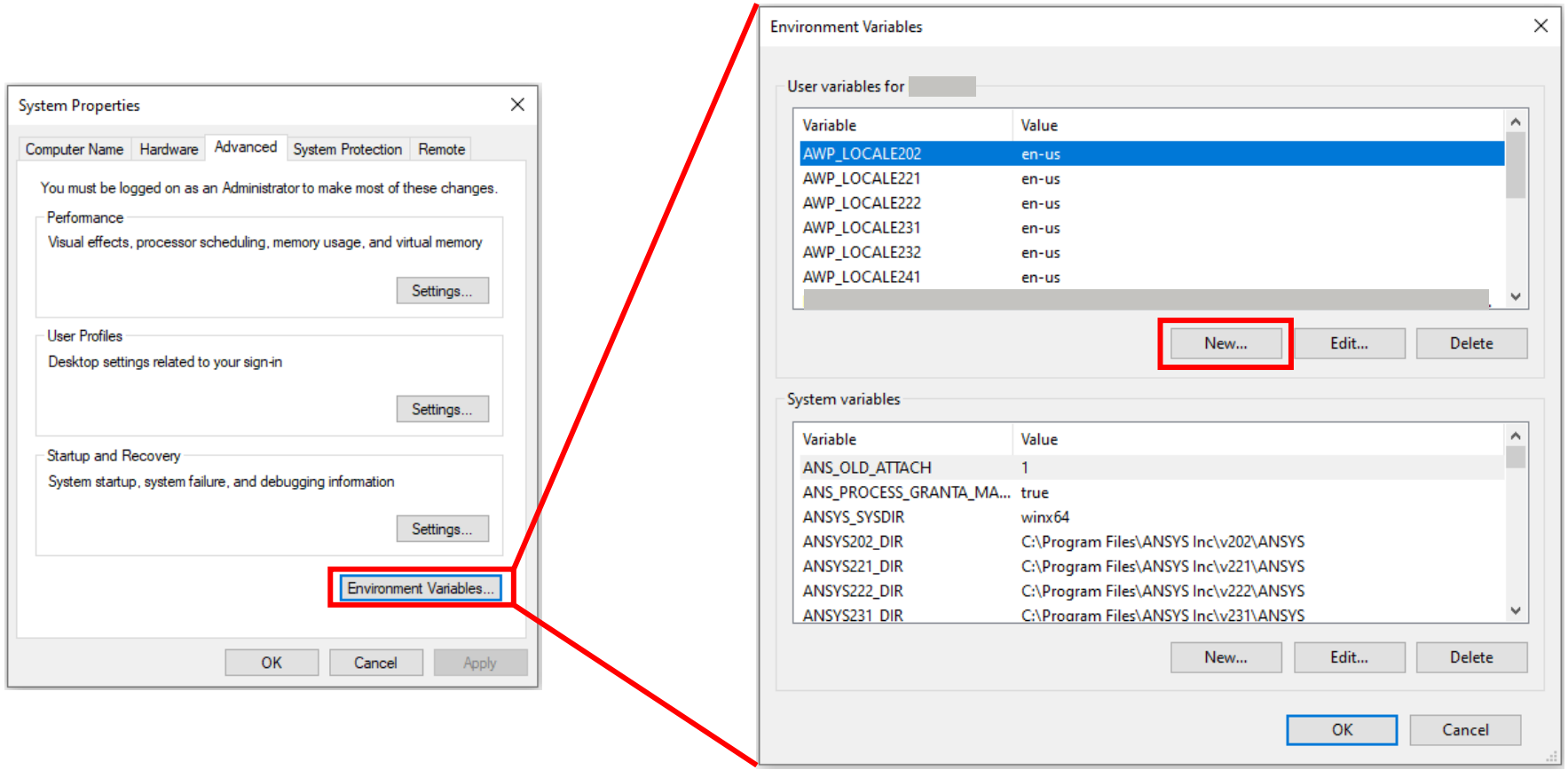
- Download the version by clicking on the link.
- Double click on the downloaded exe and agree to the terms and conditions.
- Choose the destination folder where you want to extract the latest version. After clicking Ok, you will get folder with 4 files as shown.



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Setting environment variable for LS-DYNA version

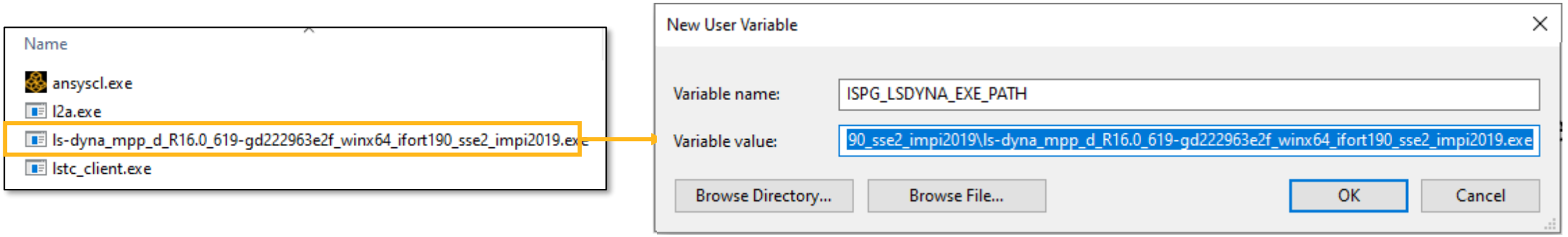
- Search for ‘Environment Variables’ on your PC settings and click on Environment Variables. In the User variables, click on ‘New..’



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Setting environment variable for LS-DYNA version

- Set the Variable name as '**ISPG_LSDYNA_EXE_PATH**'
- For the variable value, click on 'Browse File' and select the exe of the latest version of LS-DYNA from the extracted folder (slide 34).



- Click on OK.
- Restarting Workbench is necessary to recognize the new environment variable.

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How to use LS-Run

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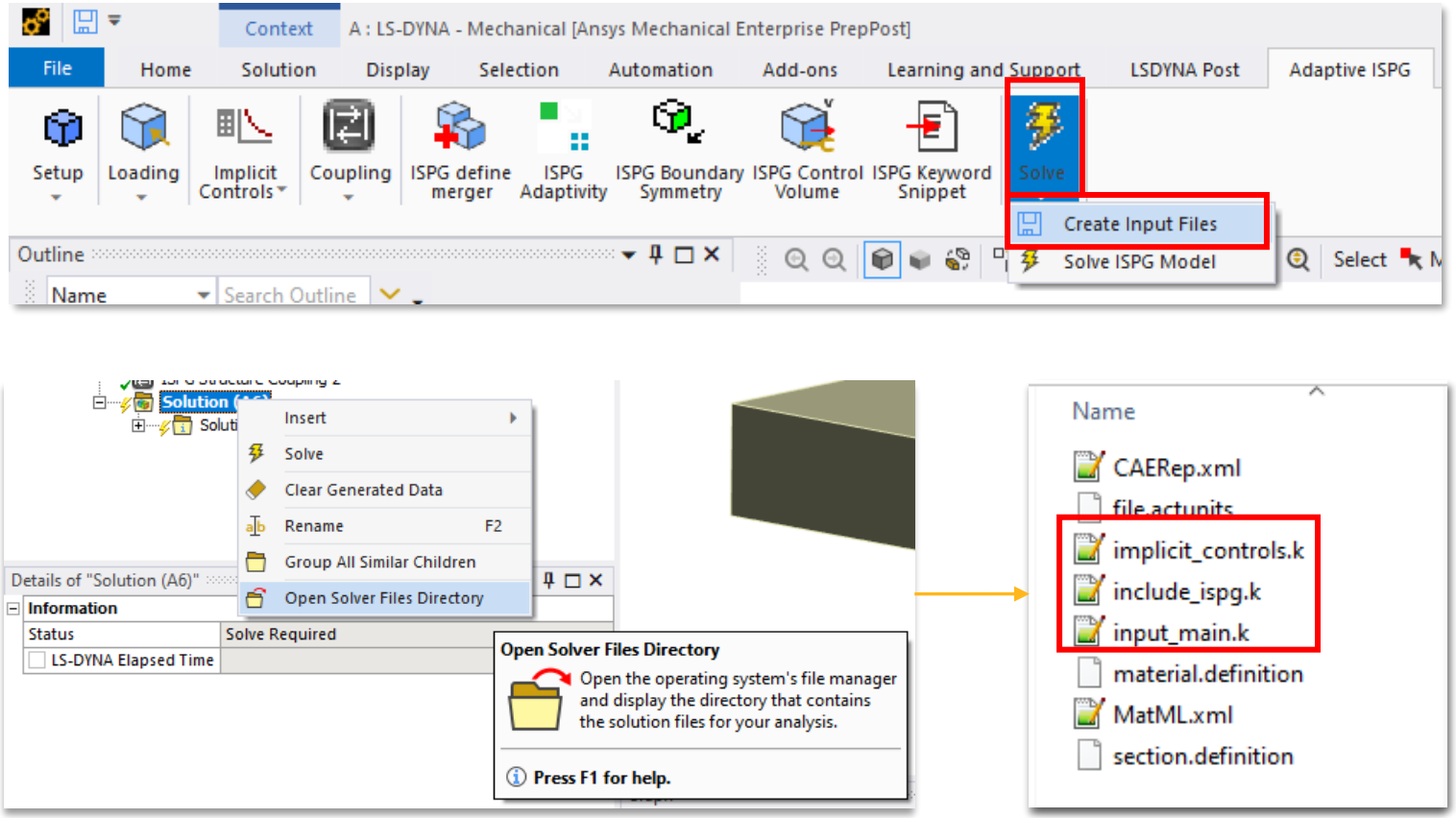
Workflow:

- Create an input deck using ‘Create input files’ from the ACT extension.
- Open LS-Run.
- Add Input File and Solver.
- Open LS-Prepost
- View results and postprocess.

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Writing input deck

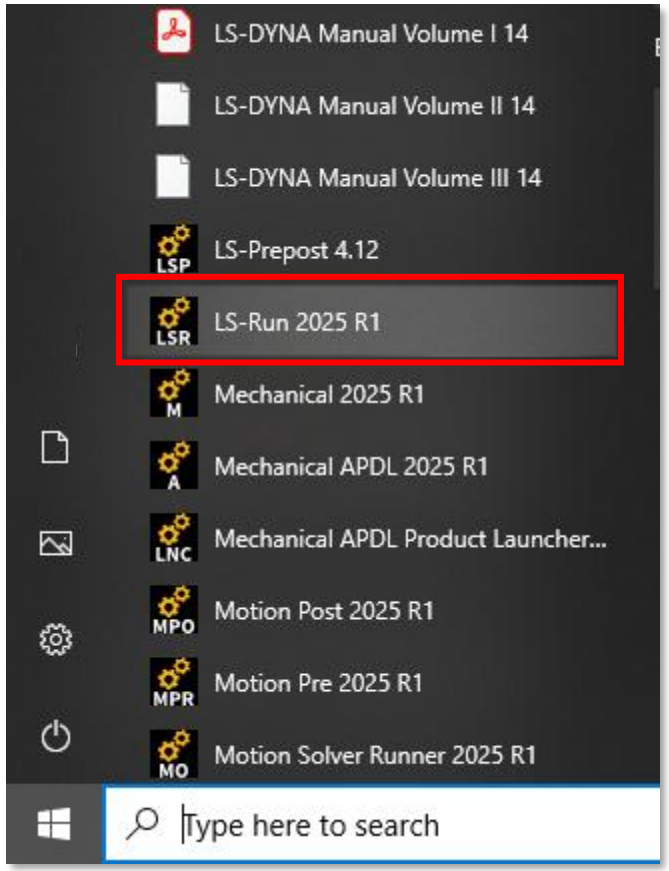
- Create an input deck using ‘Create Input Files’, which you can manually edit (if needed) and run externally, for example with LS-Run. These files are stored in the solver files directory.

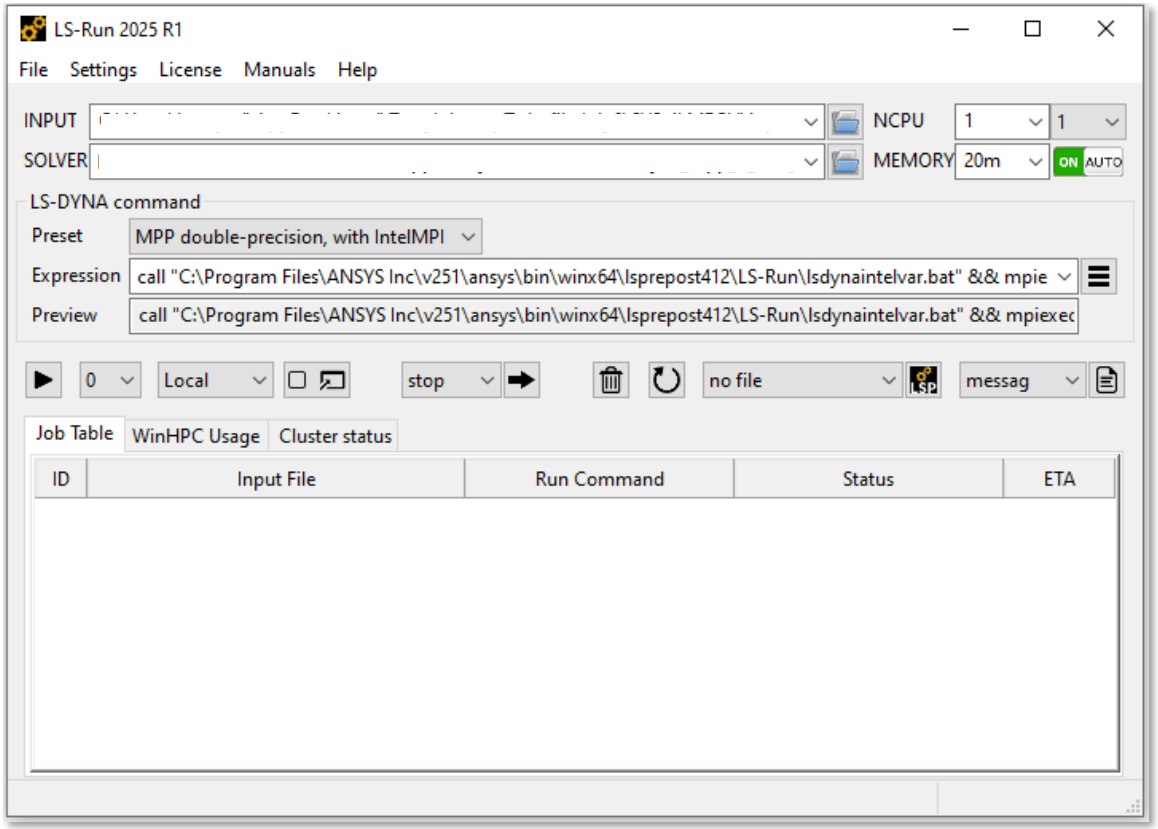


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Open LS-Run

- Double click on LS-Run to start it (Start Menu -> Under Ansys folder)

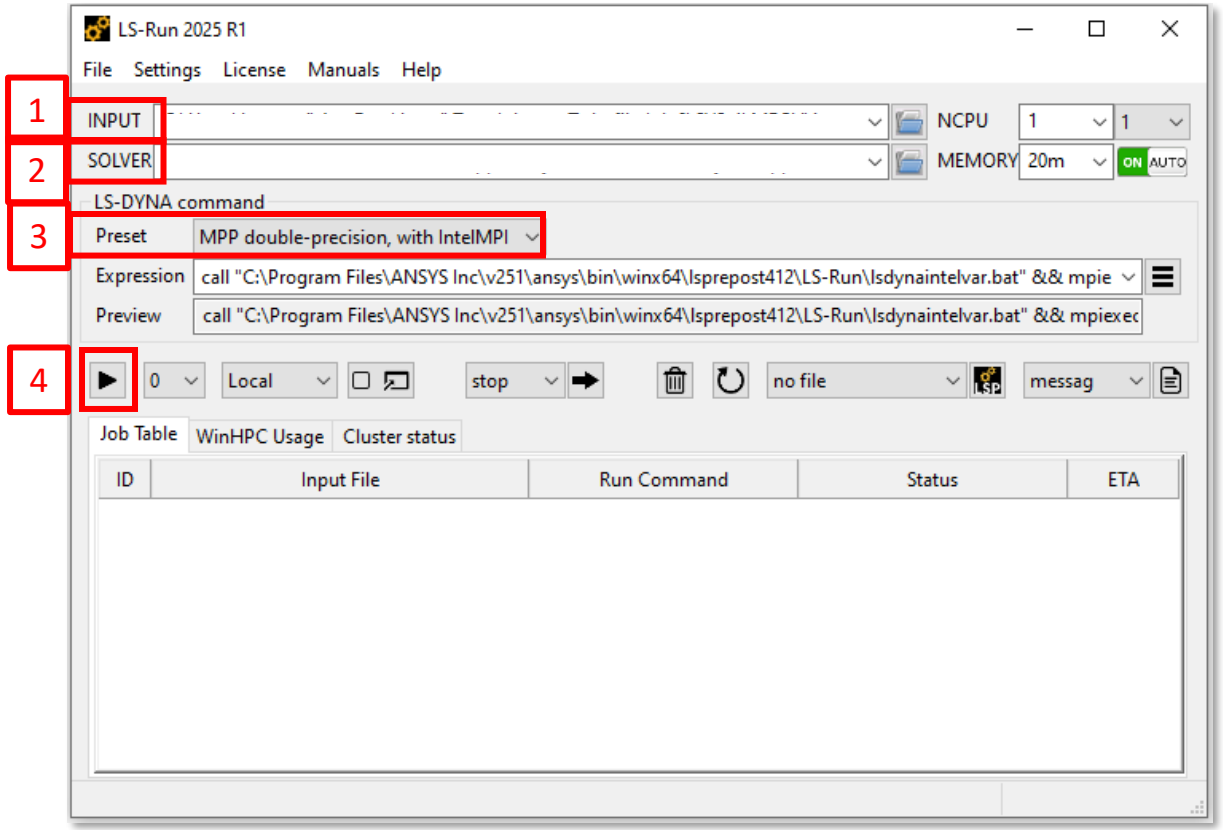




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Add input file and solver

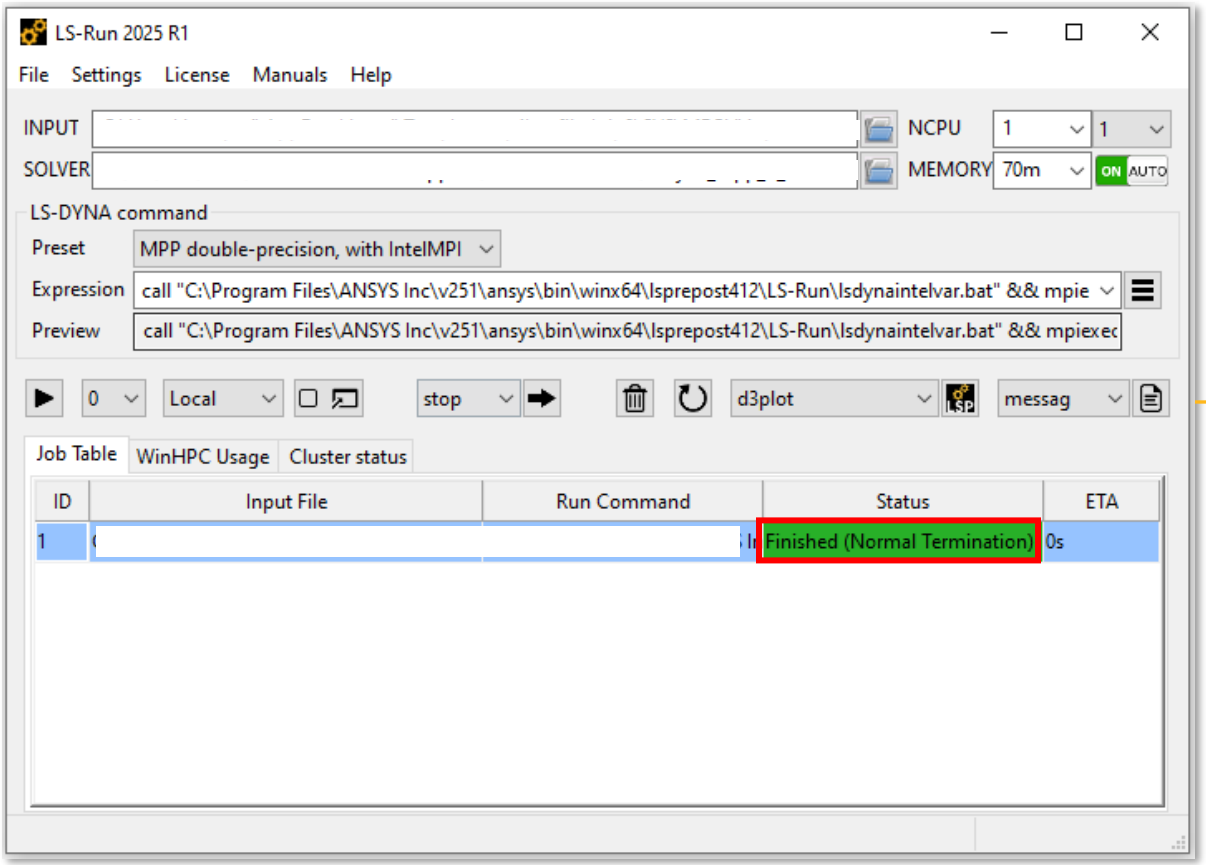
1. For INPUT, select ‘input_main.k’ file from the solver files directory (slide 39).
2. For SOLVER, select the LS-DYNA solver from the installation folder (slide 34).
3. Set the ‘Preset’ to MPP double-precision, with the MPI you have.
4. Click on ‘Add job to the local queue’ button.

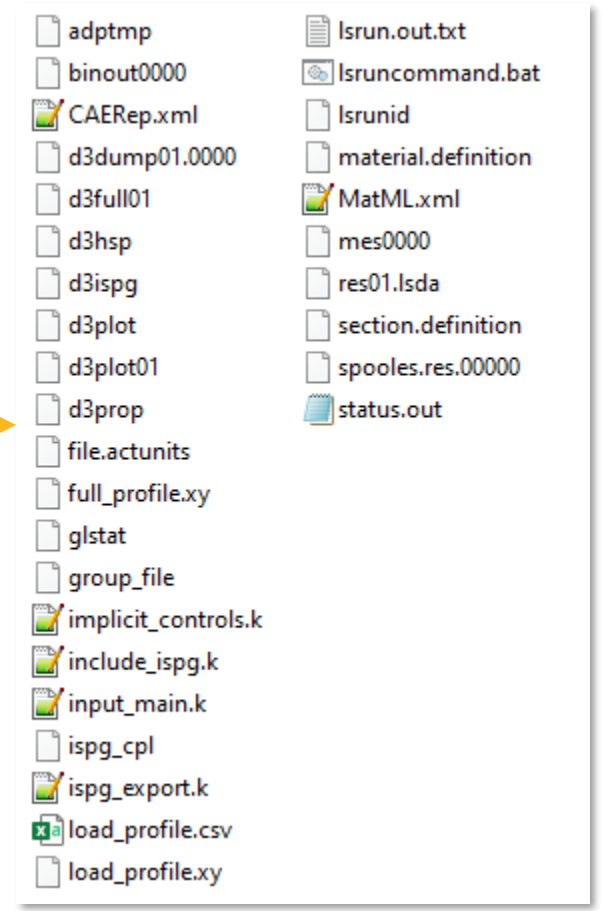


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Add input file and solver

- After a successful run, you will see multiple files added to the solver files directory in addition to the input files.

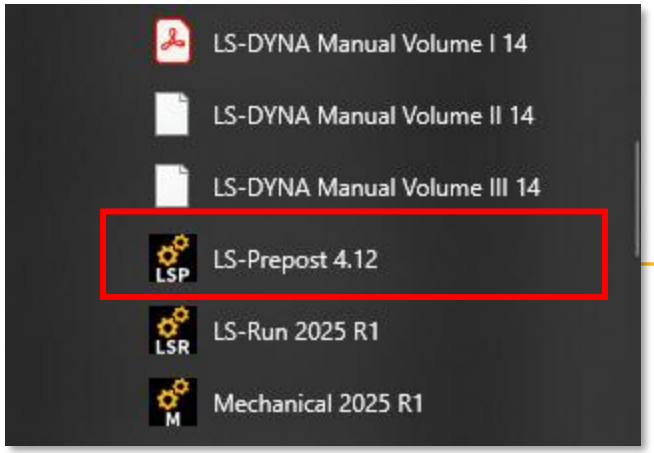


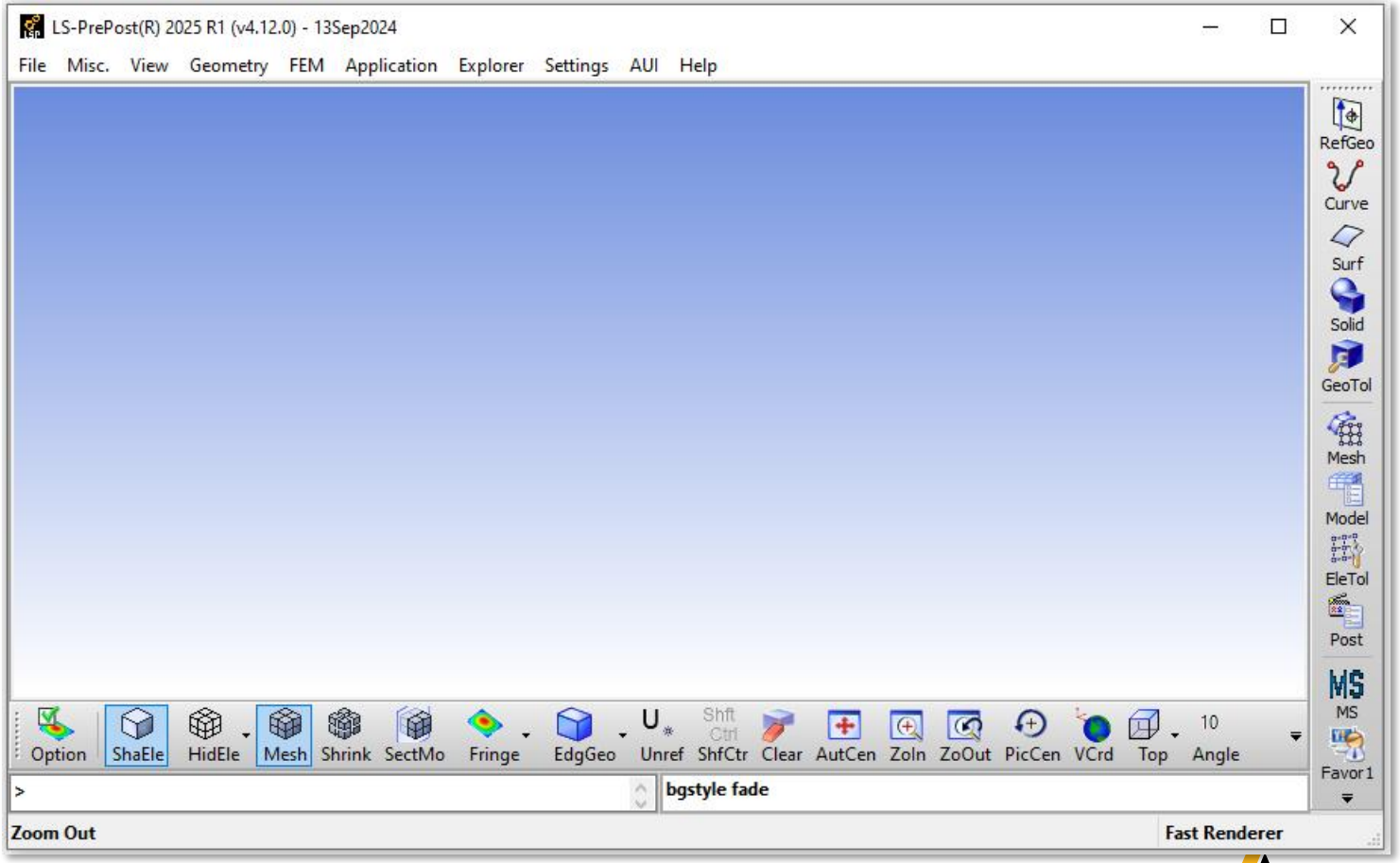


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Open LS-Prepost

- Double click on LS-Prepost to start it (Start Menu -> Under Ansys folder)

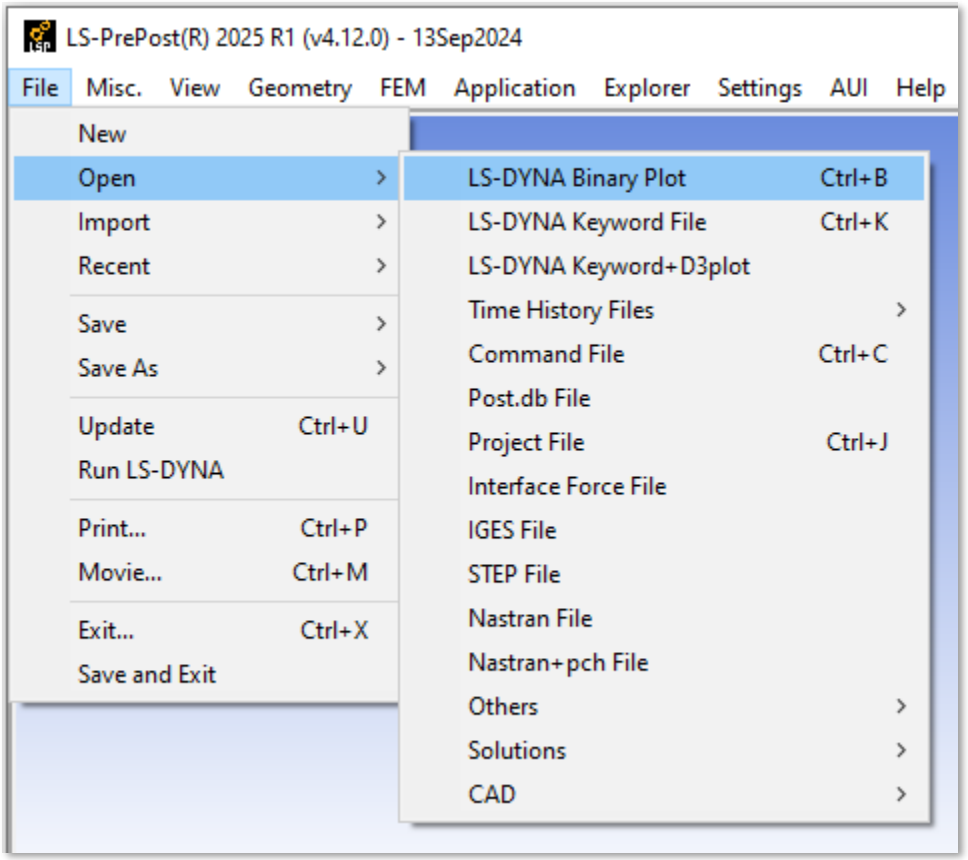


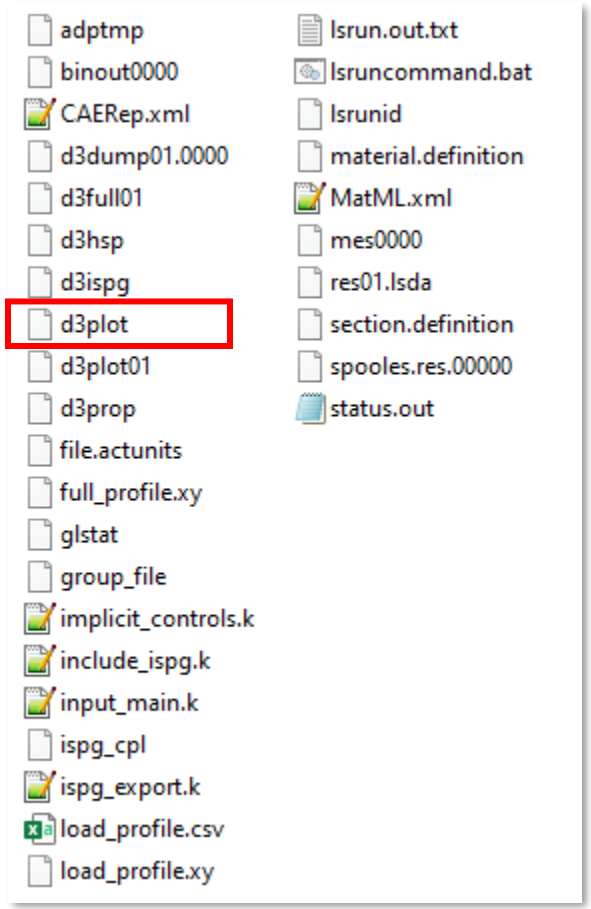


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Open LS-Prepost

- Go to File-> Open and select ‘LS-DYNA Binary Plot’. Select the ‘d3plot’ file from the solver files directory.

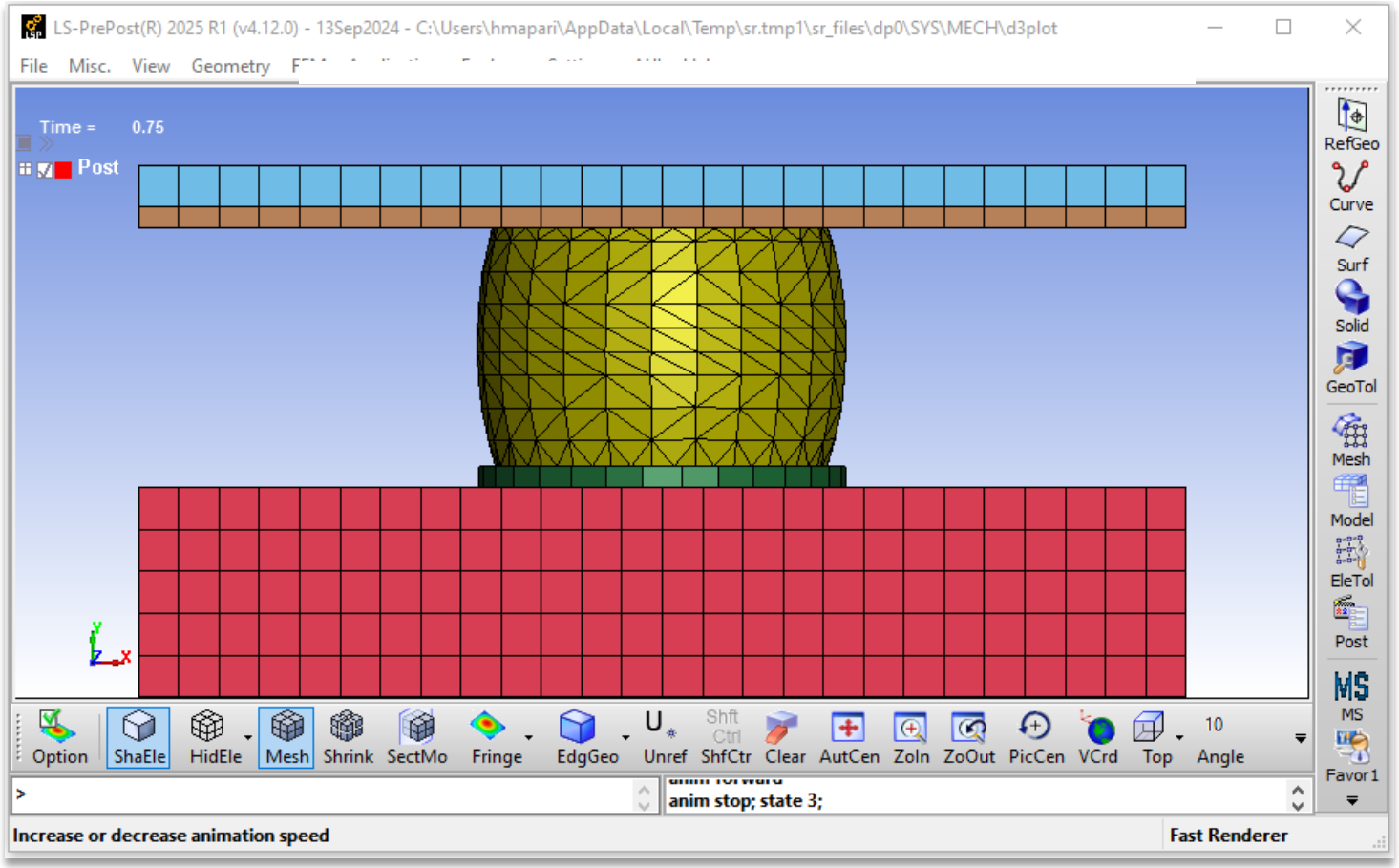




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View results and post-process

- Now, you can view the results and further post process it.



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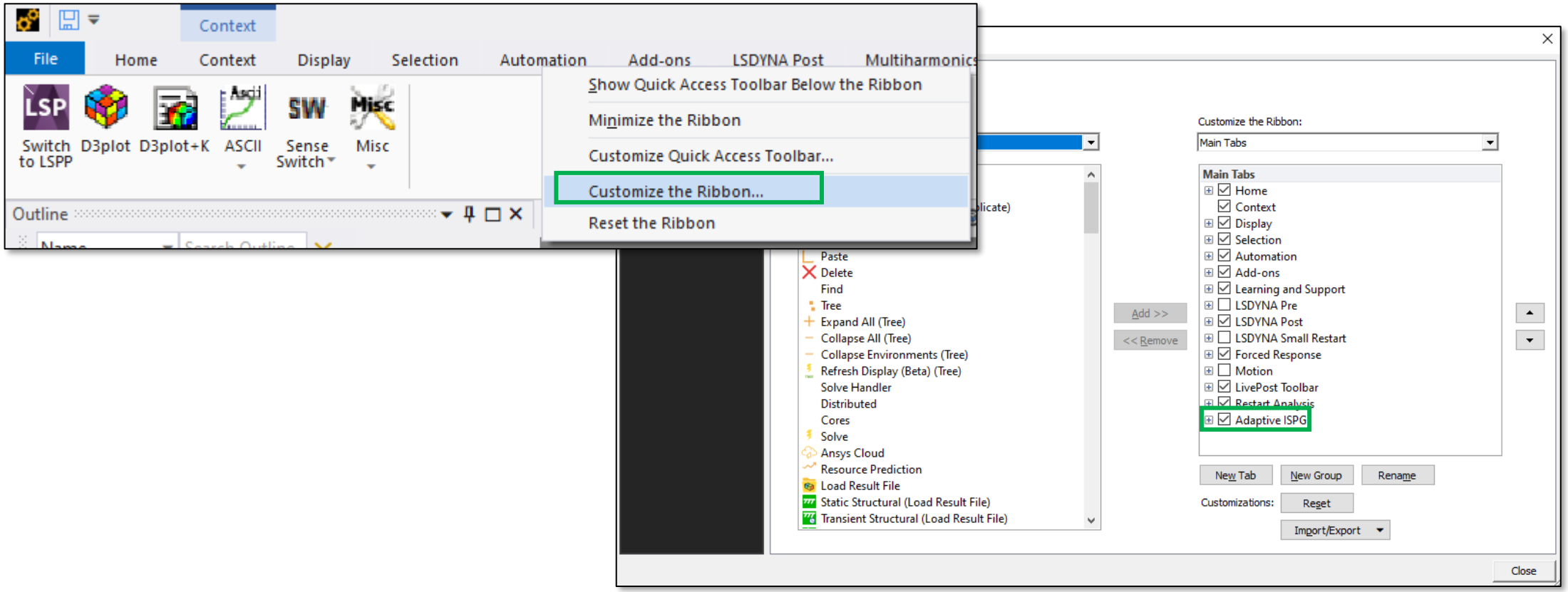
Troubleshooting

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I don’t see the ISPG App in Mechanical?

This might be because of a known issue, that Toolbar apps sometimes don’t appear even though they are loaded. If you happen to come-across this issue, please load the App manually:

Right click on the Menu-Ribbon > Customize the Ribbon… > Adaptive ISPG



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Solution is finished but no result files?

- Please make sure that you have suppressed or deleted automatically created connections.

- Make sure that all ISPG parts have tetrahedral elements.

- Check solver log “mes0000” or “d3hsp” file.

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